

Technical Report 3: XGBoost for Forecasting I-SEM Day-Ahead Market Prices

Abstract

Accurate day-ahead electricity price forecasting is crucial for efficient participation in the I-SEM Day-Ahead market, enabling better decision-making for utilities and consumers. In this study, we developed a forecasting model using eXtreme Gradient Boosting (XGBoost) to predict day-ahead electricity prices for Power NI. We structured historical I-SEM price data along with relevant external variables, incorporating engineered features such as hourly, daily, and seasonal patterns, peak hour indicators, and multiple lagged price features, including price lag at the previous hour, the same hour yesterday, and the same hour last week. Additionally, the gas-to-wind generation ratio was included to capture market dependencies. XGBoost was employed for feature extraction and selection before being applied to forecast electricity price. Our model achieved a Mean Absolute Error (MAE) of 12.03 and an R^2 score of 0.8890, demonstrating strong predictive performance. Compared to traditional forecasting methods, XGBoost exhibited superior efficiency in both computational speed and memory usage. The proposed scheme outperformed alternative approaches based on the Mean Absolute Percentage Error (MAPE) metric, highlighting its effectiveness for accurate and reliable forecasting of day-ahead electricity prices in the I-SEM market.

1 Introduction

XGBoost, or Extreme Gradient Boosting, represents a cutting-edge approach to machine learning that has garnered widespread acclaim for its exceptional performance in tackling classification and regression problems. By leveraging an ensemble of decision trees, XGBoost constructs a powerful predictive model through an iterative process that focuses on minimising errors. This iterative refinement, driven by gradient descent optimisation (the ability of the model to find the answer with the least error quickly), enables XGBoost to continually enhance its predictive accuracy by strategically updating the parameters of the decision trees. Moreover, XGBoost incorporates regularisation techniques, or more simply, techniques to combat overfitting, ensuring the model's ability to generalise well to unseen data. The combination of ensemble learning, gradient descent optimisation, and regularisation makes XGBoost a formidable tool for data scientists and practitioners seeking highly accurate and efficient solutions

to complex predictive modelling challenges.

The advantages of using XGBoost for day-ahead electricity price forecasting in the I-SEM market include:

- **High Predictive Accuracy:** XGBoost consistently outperforms traditional machine learning models by leveraging gradient boosting to iteratively refine predictions, reducing forecasting errors.
- **Efficiency in Training and Computation:** Unlike deep learning models, which require extensive computational resources, XGBoost is optimised for speed and memory efficiency, making it suitable for real-time forecasting applications.
- **Handling of Missing Data:** XGBoost can inherently process missing values by learning the best direction to take when encountering gaps in data, reducing the need for extensive data preprocessing.
- **Feature Importance Analysis:** The model provides insights into the most influential factors affecting electricity price fluctuations, enhancing interpretability and decision-making.
- **Robustness to Overfitting:** Regularisation techniques such as L1 (Lasso) and L2 (Ridge) ensure that the model generalises well, preventing excessive sensitivity to noise in the data.
- **Adaptability to Time Series Forecasting:** While traditionally used for classification and regression tasks, XGBoost can be effectively adapted for time series forecasting through lag-based feature engineering and rolling window techniques.

Given its ability to handle complex dependencies, optimise feature selection, and prevent overfitting, XGBoost is an ideal choice for forecasting day-ahead electricity prices in the I-SEM market.

2 Mathematical Foundation of XGBoost

XGBoost optimises the following objective function:

$$\mathcal{L}(\theta) = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (1)$$

where $l(y_i, \hat{y}_i)$ is the loss function, typically squared error for regression:

$$l(y_i, \hat{y}_i) = (y_i - \hat{y}_i)^2 \quad (2)$$

The regularisation term $\Omega(f)$ controls model complexity:

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_j w_j^2 \quad (3)$$

where T is the number of leaves, w_j are leaf weights, and λ is a regularisation parameter.

2.1 Gradient Boosting Update Rule

At each boosting step, XGBoost updates predictions using gradient and Hessian values:

$$g_i = \frac{\partial l(y_i, \hat{y}_i)}{\partial \hat{y}_i}, \quad h_i = \frac{\partial^2 l(y_i, \hat{y}_i)}{\partial \hat{y}_i^2} \quad (4)$$

New trees are fit to the gradient g_i and Hessian h_i , improving predictions iteratively.

2.2 Feature Importance Calculation

Feature importance is computed based on the contribution of each feature to reducing loss:

$$Gain = \sum_{splits} (loss_{before} - loss_{after}) \quad (5)$$

This helps in identifying the most influential factors affecting electricity price fluctuations.

3 Formulation of the XGBoost Model

3.1 Data Preprocessing and Feature Engineering

The XGBoost model we aim to build will take raw electricity price data and prepare it for forecasting by creating relevant features. The data will consist of the most recent 12 months of data, sampled at 30-minute intervals. The data preprocessing steps begin with loading the CSV file and converting the timestamps to a datetime index. We will then filter the data to focus on the last 12 months using a specified cutoff date.

For feature engineering, several types of features will be created. Time-related features will include the hour of the day, day of the week, month, whether it is the weekend, quarter of the day, and peak hours. External features such as the 'DemandForecast-DAM', 'WindForecast-DAM', gas prices, and carbon prices will also be included. Lag features, such as the previous period, the same hour yesterday, and the same hour last week, will be used to capture temporal dependencies. Additionally, a calculated feature, the 'Gas_to_Wind_Ratio', will be introduced, which represents the ratio of gas prices to the wind forecast.

Further feature engineering will be done by determining the points per day, automatically adjusting for either 30-minute (48 points) or hourly (24 points) data. Peak hours will be defined as the hours between 7:00 and 22:00, representing the periods of peak demand. The relationship between gas prices and wind generation will be captured through the 'Gas_to_Wind_Ratio' feature. Features can be added by extending the 'features' list, while time windows can be adjusted by modifying the cutoff date. Finally, lag periods can be fine-tuned by changing the shift values to better capture the temporal dynamics of the data.

3.2 Model Training

In this step, we train the XGBoost regressor model on 80% of the most recent 12 months of data and evaluate its performance on the remaining 20%. The training data consists of 14,055 points, spanning from January 20, 2024, to November 8, 2024, while the testing data includes 3,514 points, ranging from November 8, 2024, to January 20, 2025. The dataset is split as follows:

```
# Train/test split
train_size = int(len(df_recent) * 0.8)
train = df_recent.iloc[:train_size]
test = df_recent.iloc[train_size:]
```

We define the features and target variable for training and testing, where the target variable is the ISEM DA Price, and the features consist of the various time-related, external, lag, and calculated features prepared in the previous step.

```
X_train = train[features]
y_train = train['ISEM-DA-Price']
X_test = test[features]
y_test = test['ISEM-DA-Price']
```

The XGBoost model is trained using the following hyperparameters: 200 trees, a learning rate of 0.05, and a maximum tree depth of 6. We use squared error as the objective function to minimise. The training process begins with fitting the model to the training data:

```
xgb_model.fit(X_train, y_train)
```

Once the model is trained, predictions are generated for both the training and test sets, and several performance metrics are calculated for the test set, including RMSE, MAE, sMAPE, and R^2 score:

```
train_preds = xgb_model.predict(X_train)
test_preds = xgb_model.predict(X_test)
```

The model performance on the test set is summarised as follows:

- RMSE: 22.04
- MAE: 12.03

- sMAPE: 13.68%
- R^2 Score: 0.8890

Additionally, the model's performance on negative prices is assessed, and an RMSE of 22.99 is observed for this specific subset.

Next, we analyse the feature importance to understand which features contribute most to the model's predictions. The top 5 most important features are as follows:

- **Price_Lag1:** 0.5954
- **Peak_Hour:** 0.0923
- **Gas_to_Wind_Ratio:** 0.0731
- **WindForecast-DAM:** 0.0683
- **DemandForecast-DAM:** 0.0496

3.3 Model Evaluation and Visualisation

After training the XGBoost model, the next step is to evaluate its performance visually by comparing the predicted electricity prices against the actual values in the test set. Figure 1 provides a clear view of the model's forecasting ability, with training data, test data, and the model's predictions.

The plot includes three key components:

- **Training Data (Blue Line):** The actual electricity prices from the training dataset, spanning the first 80% of the time period used for training the model. This line provides the reference for the model's learning process.
- **Test Data (Green Line):** The actual electricity prices from the test dataset, representing the remaining 20% of the data. This is used to evaluate the out-of-sample performance of the model.
- **XGBoost Forecast (Red Dashed Line):** The electricity price predictions made by the XGBoost model on the test dataset. These predictions are overlaid on the actual test data to assess how well the model performs in forecasting future prices.

The key observations from Figure 1 include:

- The vertical dashed line marks the split point between the training and test data, providing a clear indication of when the model begins forecasting.
- By comparing the red dashed line (predictions) to the green line (test data), we can assess how accurately the model captures the patterns in the test set.

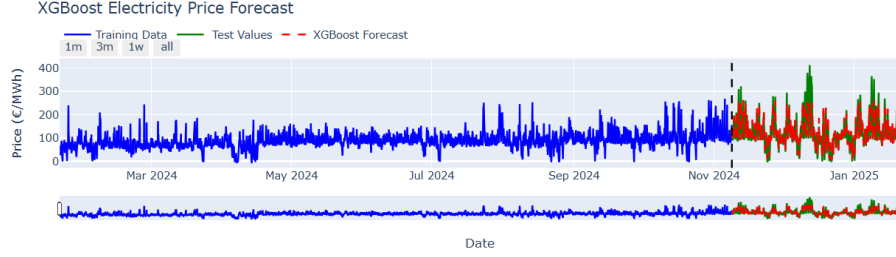


Figure 1: XGBoost Electricity Price Forecast illustrating the model’s forecasting performance, including training data, test data, and predicted values.

- The model’s forecasted values show a generally strong alignment with the actual test data, demonstrating the effectiveness of the XGBoost model in predicting electricity prices.

An interactive range selector at the bottom of the plot allows for zooming in on specific time periods. Users can select a 1-month, 3-month, or 1-week window to examine the model’s performance in finer detail. This feature provides flexibility in analysing the forecast over various time horizons.

3.4 Forecast

The forecast is set up by defining the forecast period and constructing a time index for the next 24 hours. Several features are calculated and added to the forecast DataFrame, including time-based features (e.g., hour, day of the week, month), lag features (e.g., price lag from the same hour yesterday and the same hour last week), and external variables like fuel prices and wind forecasts. These features mirror those used during model training to ensure consistency between the forecast and the model’s inputs.

The trained XGBoost model is then employed to generate predicted prices for the forecast period. Additionally, actual electricity prices from the same period in the previous week are retrieved to evaluate the model’s predictive accuracy. This historical data, along with the forecasted prices, is visualised in Figure 2, allowing a comparison between actual prices, predicted values for the previous week, and the forecast for the next day.

The graph in Figure 2 illustrates the performance of the XGBoost model in forecasting I-SEM Day-Ahead Market Prices over the past week, alongside a next-day price forecast. The visualisation compares three distinct data series: *Actual Prices* represented by the green line, *Model Predictions* shown as a red dashed line, and the *Next Day Forecast* depicted by the blue line.

The model effectively captures the general trend of market prices throughout the week, accurately following the typical daily fluctuations. However, it occasionally underpredicts sharp spikes, most notably around January 20th, where the model fails to capture a sudden peak in prices.

The next-day forecast, saved in the file `data/xgboost/xgboost_forecast.csv`, predicts a price range between €105.26/MWh at 22:00 and €157.82/MWh at 07:00, with an average of €131.87/MWh. While the model demonstrates strong overall performance in tracking standard patterns, its tendency to underpredict extreme values suggests potential areas for improvement. Enhancing the model through feature engineering, incorporating additional external data sources, or using hybrid modeling approaches may help to improve predictive accuracy, particularly for peak values.

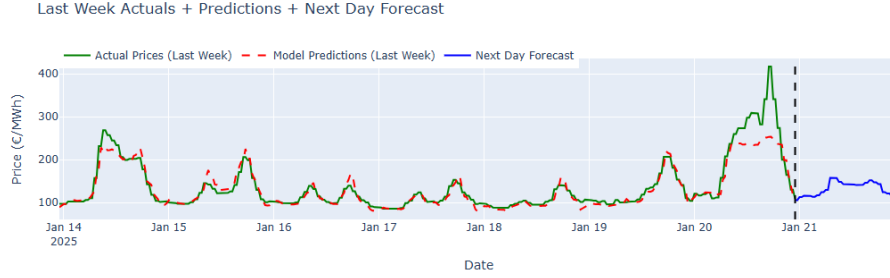


Figure 2: Forecast of electricity prices: Last week’s actual prices, model predictions, and next day’s forecast.

4 Conclusion

The XGBoost model developed for forecasting I-SEM Day-Ahead Market Prices demonstrates strong performance in predicting general price trends. By leveraging various features, including time-based metrics, lag features, and external variables such as fuel prices and wind forecasts, the model achieves a **MAE of 12.** and an **R^2 value of 0.8890**. These metrics indicate that the model has a high degree of predictive accuracy, effectively modelling the underlying patterns in the dataset.

However, the model occasionally underpredicts sharp price spikes, as evidenced by the noticeable discrepancy around January 20th. This limitation suggests that the model could benefit from further refinement, particularly through enhanced feature engineering or the incorporation of additional external data sources to better capture anomalous events. Additionally, implementing hybrid modeling techniques may further improve the model’s ability to forecast extreme values accurately.

The results of this forecasting exercise provide valuable insights for participants in the I-SEM market, allowing for more informed decision-making in response to predicted price fluctuations. Future work will focus on enhancing the model’s robustness by improving its responsiveness to sudden price peaks, thereby further increasing its practical applicability for day-ahead electricity price forecasting.